

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Indicative content: Both cans the same distance [5 cm] from the heater <i>If one can is closer to the heater it will be heated more than the other</i> Same volume of water [200 cm³] in both cans <i>If different volumes used, the temperature rise would be different</i> Same starting temperature [8 °C] of water <i>Different starting temperatures of water will affect final temperature</i> Same heater used to heat both cans <i>Both cans heated by the same power output</i> Both cans have lids <i>Reduces convection from both cans</i> Both cans contain water <i>If different liquids used, the temperature rise would be different</i> Both cans identical [size / material], apart from colour <i>Different sizes / materials may absorb radiation differently</i> Both cans heated for the same time [8 min] <i>Energy from the heater same for both cans</i> 5–6 marks Identify 3 controlled variables with correct explanations. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	3	3		6		6